1 intro

Day to day variation causes error: I.e. day to day temperature fluctuations might change the power available for a laser? (check this?).

One standard deviation is $\frac{1}{\sqrt{e}}$ away - about 63%.

Imagine taking 5 data points to get $\bar{y}$. If a million people did 5 trials, the $\bar{y}$ has its own distribution.
As $N \to \infty$, the $\bar{y}$ distribution has a mean μ (true mean) and standard deviation $\frac{\sigma}{\sqrt{N}}$, which is the Standard Error of Measurement.

Goal of lecture

Given N trials $\{y_1, y_2, \ldots, y_n\}$ drawn independently, each according to probability distribution $P(y)$ with mean μ and variance σ^2 unknown to us, estimate μ and attach an uncertainty to our estimate of μ.

Sample mean is an unbiased estimate of μ

$$\bar{y} = \frac{1}{N}\sum_{n=1}^{N} y_n$$

The sample variance is an unbiased estimate of the variance σ^2

$$s^2 = \frac{1}{N-1}\sum_{n=0}^{N}(y_n - \bar{y})^2$$

And the standard error of the mean gives the uncertainty of $\bar{y}$ as an estimate of μ. This is the distribution that the $\bar{y}$ s follow, with a standard deviation well approximated by
$s_{\bar{y}} = \frac{s}{\sqrt{N}}$.

SEM is the size of error bars for each data points.

Mean and variances

$$\begin{aligned}
&\text{total probability is 1 (normalization)} \\
&\int_{-\infty}^{\infty} P(y)dy = 1 \\
&\text{mean:} \\
&\langle y \rangle \equiv \int_{-\infty}^{\infty} yP(y)dy = \mu \\
&\text{variance:} \\
&\langle (y-\mu)^2 \rangle \equiv \int_{-\infty}^{\infty} (y-\mu)^2 P(y)dy = \sigma^2 \\
&\text{Standard Dev}: \\
&\sqrt{\sigma^2}
\end{aligned}$$

For a normal distribution,

$$\int_{\mu-\sigma}^{\mu+\sigma} P(y)dy \approx 0.68$$

for 2σ it is 0.95, and 3σ is 0.997.

Warmup

y is drawn from $P(y)$, z is drawn from $P(z)$, with respective means and variance.
They are drawn independently, so $P(y,z) = P(y)P(z)$.
If we have a function $f(y,z)$, we make a linear approximation

$f(y,z) \approx f(\mu_y,\mu_z) + \frac{\partial f}{\partial y}\Big|_{\mu_y,\mu_z}(y-\mu_y) + \frac{\partial f}{\partial z}\Big|_{\mu_y,\mu_z}(z-\mu_z)$

We can propagate error.

$$\underbrace{\langle f(\mu_y,\mu_z)\rangle}_{f(\mu_y,\mu_z)} + \left\langle \frac{\partial f}{\partial y}\Big|_{\mu_y,\mu_z}(y-\mu_y)\right\rangle + \left\langle \frac{\partial f}{\partial z}\Big|_{\mu_y,\mu_z}(z-\mu_z)\right\rangle$$

$$\left\langle [f(y,z) - f(\mu_y,\mu_z)]^2\right\rangle = \left\langle \left[\frac{\partial f}{\partial y}\Big|_{\mu_y,\mu_z}(y-\mu_y) + \frac{\partial f}{\partial z}\Big|_{\mu_y,\mu_z}\right]^2\right\rangle \text{ which we can foil into three terms } = \left\langle \underbrace{\left(\frac{\partial f}{\partial y}\Big|_{\mu_y,\mu_z}\right)^2}_{\text{constant}} \underbrace{(y}_{\text{vari}}$$

We can calculate this last term as

$$\langle (y-\mu_y)(z-\mu_z)\rangle = \iint (y-\mu_y)(z-\mu_z)P(y,z)dydz$$

but because they are independent, $P(y,z) = P(y)P(z)$ so we have this last term as $\langle y-\mu_y\rangle \langle z-\mu_z\rangle = 0 \cdot 0$
Therefore, we have found that

$$\left\langle [f(y,z) - f(\mu_y,\mu_z)]^2\right\rangle = \left(\frac{\partial f}{\partial y}\Big|_{\mu_y,\mu_z}\right)^2 \sigma_y^2 + \left(\frac{\partial f}{\partial z}\Big|_{\mu_y,\mu_z}\right)\sigma_z^2\omega_f^2 \approx \left(\frac{\partial f}{\partial y}\right)^2\sigma_y^2 + \left(\frac{\partial f}{\partial z}\right)^2\sigma_z^2$$

Lets prove things. That for N independent trials,
$\bar{y} = \frac{1}{N}(y_1 + y_2 + \cdots + y_n)$

$$\begin{aligned}\langle \bar{y}\rangle &= \frac{1}{N}\sum \langle y_n\rangle \\ &= \frac{1}{N}N\mu = \mu\end{aligned}$$

What is the variance of $\bar{y}$?

$$\begin{aligned}\bar{y} = \frac{1}{N}(y_1 + y_2 + \cdots + y_n) &= \frac{1}{N}\sum_{n=1}^{N} y_n \\ \left\langle (y_n-\mu)^2\right\rangle &= \sigma^2 \\ \left\langle (\bar{y}-\mu^2)\right\rangle &= \left\langle \left(\frac{1}{N}\sum_{n=1}^{N} y_n\right) - \frac{N\mu}{N}\right\rangle \\ &= \left\langle \frac{1}{N^2}\left(\sum_{n=1}^{N}(y_n-\mu)\right)^2\right\rangle\end{aligned}$$

This has N terms that look like $\left\langle (y_n-\mu)^2\right\rangle = \sigma^2$, and a bunch of terms like $\langle (y_n-\mu)(y_m-\mu)\rangle = 0$
So, we have

$$\left\langle (\bar{y}-\mu)^2\right\rangle = \frac{1}{N^2}N\sigma^2 = \frac{\sigma^2}{N}$$

If we let a million people do a 5 trials experiment ($N=5$), then we see that the expectation value of the variance decreases by N - which is what leads to SEM $= \frac{\sigma}{\sqrt{N}}$.

However, if we don't know what σ is in an experiment (which is most of the time), we need to estimate σ.

We claim that the sample variance is an unbiased estimate of variance.
$\left\langle s^2\right\rangle = \sigma^2$

$$
\begin{aligned}
\left\langle s^2 \right\rangle &= \left\langle \frac{1}{N-1} \sum_{n=1}^{N} (y_n - \bar{y})^2 \right\rangle \\
&= \frac{1}{N-1} \sum_{n=1}^{N} \underbrace{\left\langle (y_n - \bar{y})^2 \right\rangle}_{\text{smaller than } \sigma^2 \text{ because } y_n \text{ helped d}}
\end{aligned}
$$

Lets do this.

$$
\begin{aligned}
\left\langle (y_n - \bar{y})^2 \right\rangle &= \left\langle [(y_n - \mu) - (\bar{y} - \mu)]^2 \right\rangle \\
&= \left\langle (y_n - \mu)^2 \right\rangle + \langle \bar{y} - \mu \rangle^2 - 2 \left\langle (y_{\bar{n}} - \mu)(\bar{y} - \mu) \right\rangle \\
&= \sigma^2 + \frac{\sigma^2}{N} - 2\xi
\end{aligned}
$$

where

$$
\xi = \langle (y_n - \mu)(\bar{y} - \mu) \rangle = \underbrace{\left\langle (y_n - \mu) \left(\frac{1}{N} \sum_{i=1}^{N} (y_i - \mu) \right) \right\rangle}_{\frac{1}{N} \times 1 \text{ term } \langle (y_n - \mu)^2 \rangle \text{ and a bunch of } 0}
$$

$$
\begin{aligned}
\left\langle (y_n - \bar{y})^2 \right\rangle &= \sigma^2 + \frac{\sigma^2}{N} - 2 \underbrace{\xi}_{\frac{\sigma^2}{N}} \\
&= \sigma^2 \left(1 - \frac{1}{N} \right) = \sigma^2 \left(\frac{N-1}{N} \right)
\end{aligned}
$$

so our best estimate of SEM is
$\mu = \bar{y} + \frac{s}{\sqrt{N}}$

$$
s = \sqrt{\frac{1}{N-1} \sum_{n=1}^{N} (y_n - \bar{y})^2}
$$